



Department of Mathematics & Natural Sciences

MAT 123: Calculus I

Spring 2026

Consultation Schedule

Will be provided by the instructor.

Instructor's Information

Dr. A H M Mahbubur Rahman

Assistant Professor of Mathematics

Department of Mathematics and Physical Sciences

BRAC University

Course Aims and Outcomes

Content in this course will be adaptive, allowing students to achieve a certain concept before moving on to the next level.

Course Contents:

Part 1:

Functions, Limits, continuity, Extreme value theorem, The Intermediate Value theorem (IVT), Differentiability, Differentiation, Applications of differentiation, Fermat's theorem Rolle's and the Mean value theorem, optimization, Taylor and Maclaurin Polynomials

Part 2:

Antiderivatives, Riemann Sums, Riemann Integral by definition via upper and lower integrable function, The Indefinite Integral, Techniques of Integration: by substitution, Integrating by parts, Trigonometric substitution, Definite Integral, Fundamental theorem of Calculus, Reduction formula, Integrating Trigonometric Functions, Integrating Rational Functions by Partial Fractions, Taylor and Maclaurin Series, Improper Integral, Gamma and Beta functions, Area between the curves, Volume by slicing, Volume by cylindrical shells

Marks Distribution

Attendance = 5%

Assignment = 10%

Quiz (best 2 out of 3) = 20%

Midterm = 30%

Final = 35%

Total = 100%

Attendance	Marks
90% and above	5
85% - 89%	4
80% - 84%	3
75% - 79%	2
70% - 74%	1
Less than 70% (Unable to sit for Final Exam)	

Lecture Plan

Week 1	Functions, Zoo of functions, Introduction to Limits with ($\varepsilon - \delta$ definition), Squeeze Theorem
--------	--

Week 2	Introduction to continuity with ($\varepsilon - \delta$ definition), Supremum, Infimum, Extreme value theorem and The Intermediate Value theorem (IVT)
Week 3	Differentiability, Techniques of Differentiation, Successive differentiation
Week 4	Limits of indeterminate forms using L' Hôpital's rule, Fermat's theorem, Rolle's and the Mean value theorem
Week 5	Maxima and Minima, Graph of Functions, Taylor and Maclaurin polynomials for single variable functions
Mid Term	
Week 6	Introduction: Antiderivatives, Riemann Sums, Riemann Integral by definition via upper and lower integrable function
Week 7	The Indefinite Integral, Techniques of Integration: by substitution, Integrating by parts, Trigonometric substitution
Week 8	The definite Integral, Fundamental theorem of Calculus, Reduction formula, Integrating Trigonometric Functions
Week 9	Integrating Rational Functions by Partial Fractions, Taylor and Maclaurin Series
Week 10	Improper Integral, Gamma and Beta functions
Week 11	Area between the curves, Volume by slicing, Volume by cylindrical shells
Final	

Administrative information and Course Requirements

- There will be no makeup quizzes.
- For makeup midterm, a student must submit application through the corresponding chair/dean of the department within an appropriate time range.
- No students will be allowed to sit for final exam if he/she misses 30% of the total classes.
- You will find all supporting documents at **slack**.
- You are strongly encouraged to drop by my office to ask questions and discuss problems.

Reference Books:

- Calculus (4th Edition) – Michael Spivak
- Calculus, Early Transcendentals (10th Edition) by Howard Anton
- Mathematical Analysis (2nd Edition) by Tom M. Apostol
- Calculus, Early Transcendentals (9th Edition) by Stewart, Clegg, Watson

Academic Integrity

Each student in this course is expected to abide by the BRAC University Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work. Also refer to plagiarism policy that will be on the web.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

If copying occurs, both the student who copied work from another student and the student who gave material to be copied will both automatically receive a zero for the quizzes and assigned work at home. Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Attendance Requirement

Attendance in a class is mandatory.

A student with class attendance below 70% must consult the instructor in order to discuss the completion of the course.

Student absent in 3 consecutive classes will require a chairperson's permission to attend the following classes.

Student unable to attend classes for known reason, must apply to the Chairperson, with copies to the course teacher and the Registrar's Office, mentioning the dates and reasons for absence. If it is not possible to inform it in advance, the Chairperson must be informed by sending an application through messenger, post, fax, or email.

Source: <<https://www.bracu.ac.bd/academics/policies-and-procedures>>